Economic Case for Trade

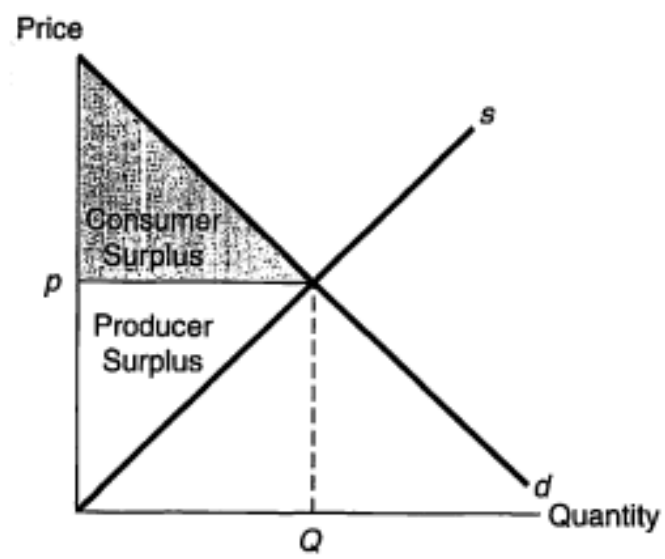
Week 11

- The Economic Case for Trade
  - Partial equilibrium
  - General equilibrium
  - Absolute advantage
  - Comparative advantage
  - Factor endowments
- Politics of trade agreement
  - Enforcing agreement

Economic Case for Trade

Partial Equilibrium

- What happens in the market for a single commodity.
- Under autarky (the country does not trade)



A. Autarky

Figure 3.1 Trade in Partial Equilibrium.

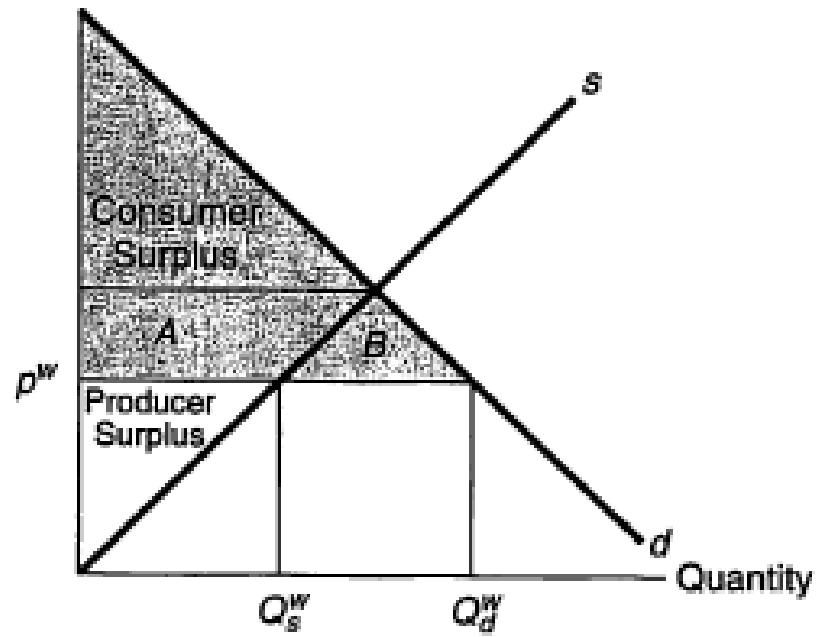
Partial Equilibrium: Free Trade

- Individual is a price taker
- Domestic economy a country is price taker (cannot influence international supply and demand)

- Domestic shirt market takes world price (p^w)

- Domestic
  - Consumers buy Q_d^w
  - Producers produce Q_s^w
  - $Q_d^w - Q_s^w$ = Imports

- Consumer surplus increase by A+B
- Producer surplus decrease by A
- Aggregate social welfare is higher



B. Free trade

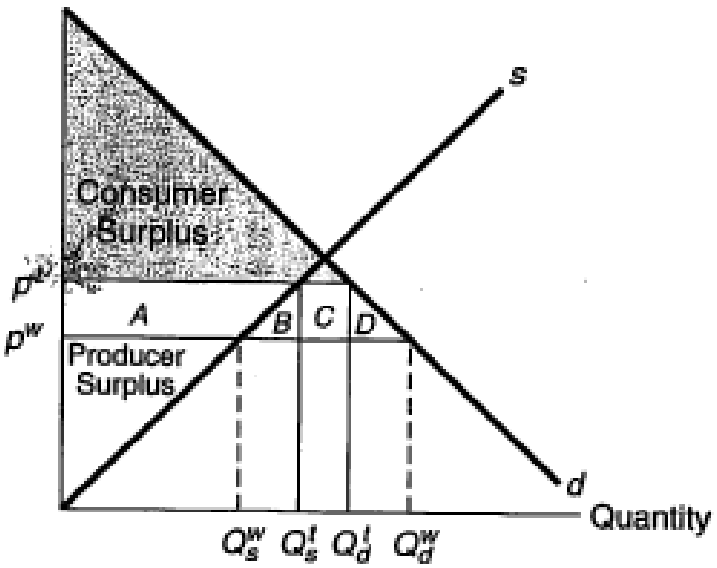
Partial Equilibrium: Tariff

- Government put tariff making the price higher (p^t)

- Consumption Q_d^t
- Supply Q_s^t
- Import decrease by $(Q_d^w-Q_s^w)-(Q_d^t-Q_s^t)$

- Consumer surplus decrease by A+B+C+D
- Producer surplus increase by A

- Tariff transfers welfare from consumers to producers (A)
- Government tariff revenue (C)
- Deadweight Loss (B+D) - efficiency losses
  - D – consumption distortion
  - B – production distortion



C. The impact of a tariff

Table 3.1 The Costs of Protection in the United States (Millions of Dollars)

Product Category	Loss of Consumer Surplus (A + B + C + D)	Gain in Producer Surplus (A)	Tariff Revenue or Quota Rents (C)	Deadweight Loss (B + D)
Tariffs				
Ball Bearings	64	13	50	1
Canned Tuna	309	127	172	10
Ceramic Articles	102	18	81	2
Ceramic Tiles	139	45	92	2
Frozen Concentrated Orange Juice	281	101	145	35
Glassware	266	162	95	9
Luggage	211	16	169	26
Polyethylene Resins	176	95	60	20
Rubber Footwear	208	55	141	12
Women's Footwear, except athletic	376	70	295	11
Women's Handbags	148	16	119	13
Total	2,280	718	1,419	141
Voluntary Export Restraints				
Apparel	21,158	9,901	8,956	2,301
Textiles	3,274	1,749	1,345	181
Machine Tools	542	157	350	35
Total	24,974	11,807	10,651	2,517

Source: Hufbauer and Elliot 1994.

Quota Rents: above-market returns created by the quota Transferred to foreign producers

1% of US GNP
$13 per person

But

$170,000 of consumer welfare decrease
$54,000 efficiency losses
per job created per year

General Equilibrium

- 2 countries
- 2 products
- 2 factors of production

- Production Possibility Frontier (PPF)
  - Countries are endowed with finite factors of production

- Slope of line AB: marginal rate of transformation
  - 1 computer:3shirts
  - Straight – constant opportunity costs
  - Curved – increasing opportunity costs; decreasing marginal returns
  - Curved – decreasing opportunity costs (economy of scale or specialization)

- U – consumption indifference curves
  - consumer utility
  - Ua - a, b, c (same utility)
  - Ub – d, e, f (same utility) higher utility than Ua

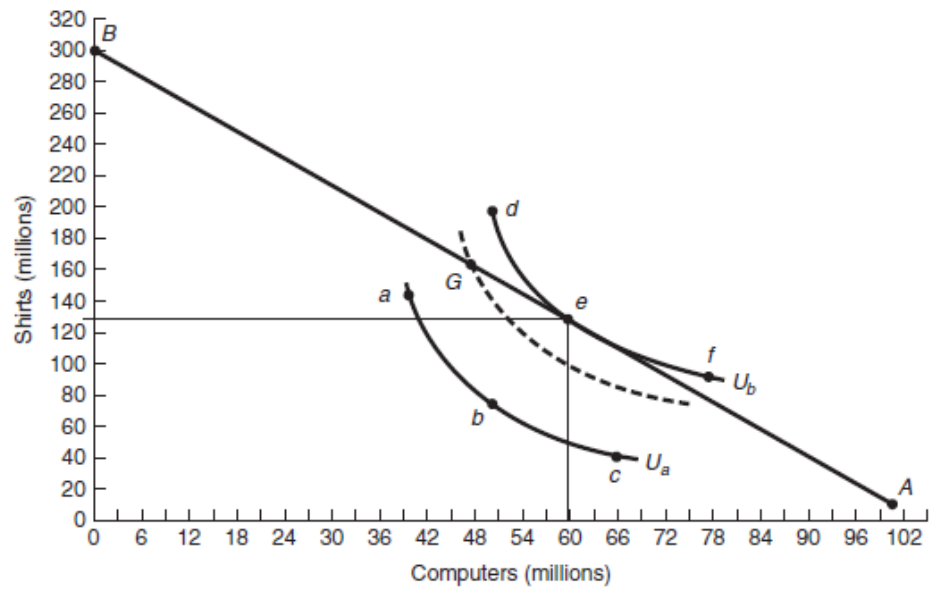


FIGURE 3.1
U.S. Production Possibility Frontier

Indifference curves

slope downwards
decreasing slope
(decreasing marginal utility)
community indifference curves
(aggregate utility for all consumers)

General Equilibrium: US Autarky

- Production will happen where the marginal rate of transformation (the slope of the PPF) = the marginal rate of substitution (the slope of the indifference curve)
  - e
  - Consumers cannot achieve higher utility

- Under autarky
  - 60 million computers
  - 120 million shirts

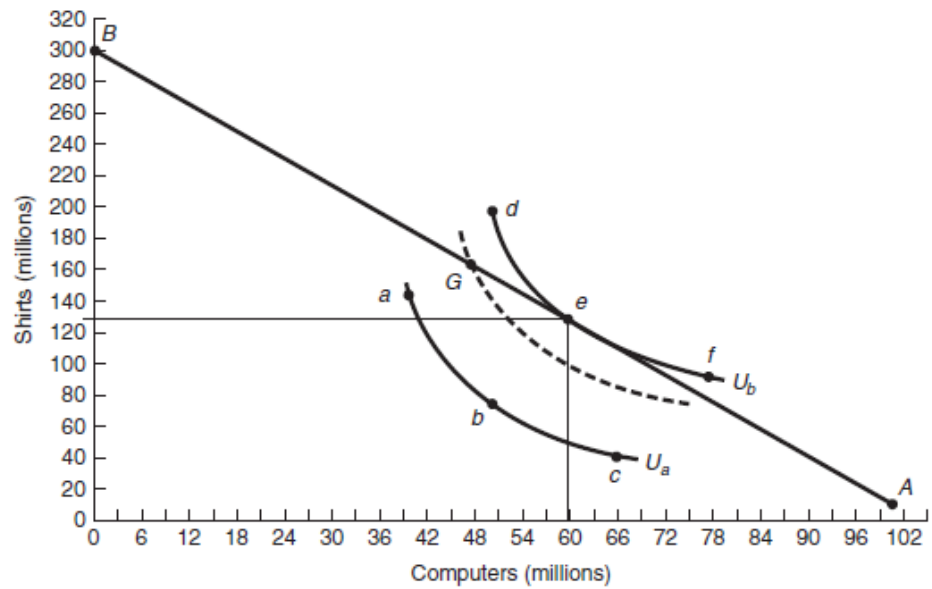


FIGURE 3.1
U.S. Production Possibility Frontier

General Equilibrium: China

- Marginal rate of transformation
  - 20 computers: 400 shirts
- Under autarky
  - 13 million computers
  - 140 million shirts

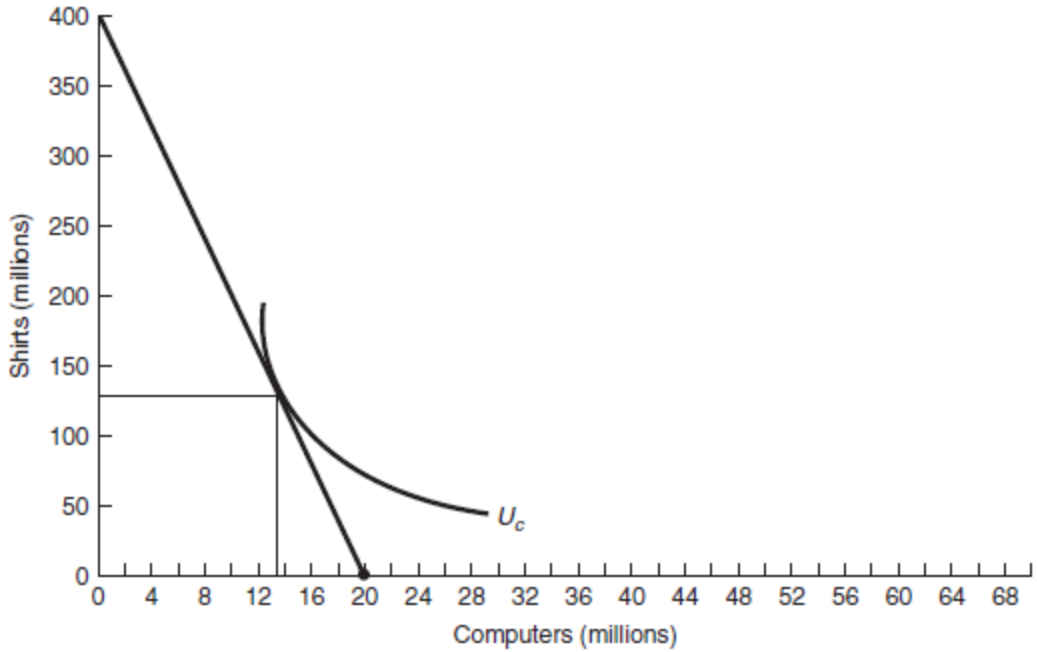


FIGURE 3.2
China's Production Possibility Frontier

General Equilibrium: trade

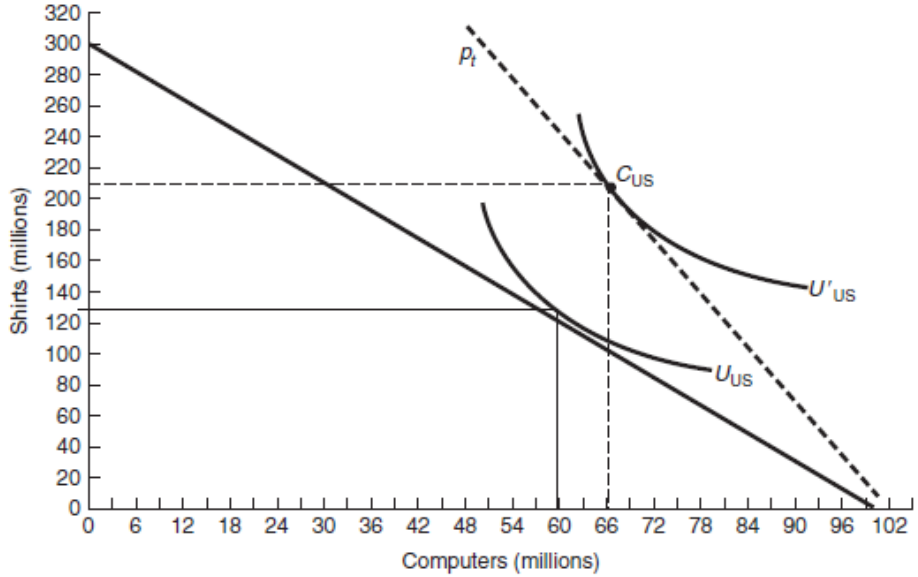
- US specialize on computer
- China specialize on shirts
- Trade at 6 shirts per computer

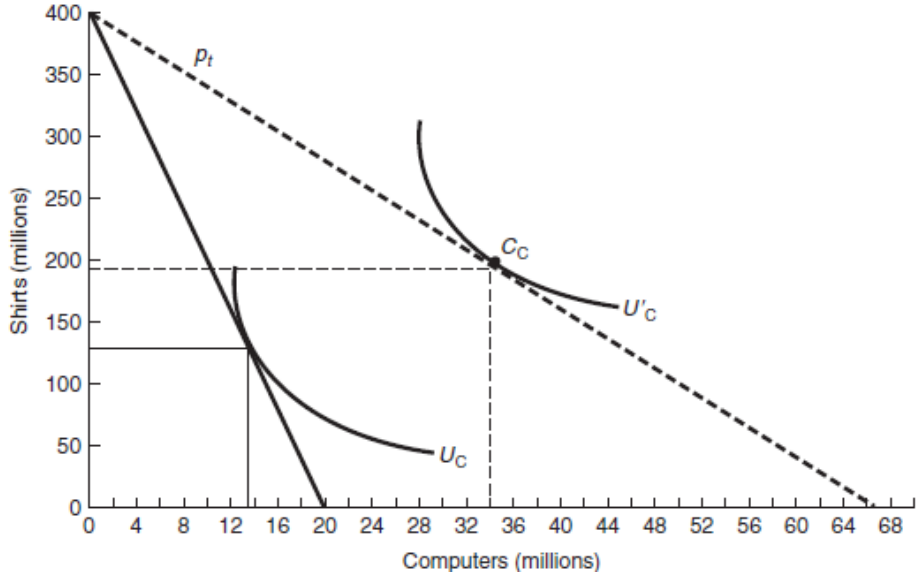
- New PPF p_t
- New equilibrium at C_{us} and C_c

- Equilibrium consumption
  - US
    - 60 m → 75 m computers
    - 120 m → 150 m shirts
  - China
    - 13 m → 25m computer
    - 140 m → 250m shirts

- Every country gains by specializing in goods it produces relatively well and trading them for goods it produces relatively less well.

- US gains not because it can produce computer more cheaply than China but it can acquire more shirts per computers by trading with China





Absolute advantage

- Productivity per unit of hour

	Computers	Cameras
Japan	8	4
US	4	8

- Japan has absolute advantage in computers
- US has absolute advantage in cameras
- If both countries put 100 hours for the production of each goods. . .

	Computers	Cameras
Japan	800	400
US	400	800

- If Japan specializes on computers and put 120 hours on computer production while US puts 120 hours on camera production. . .

- Before specialization

	Cameras	Computers
Japan	800	400
US	400	800

- After specialization

	Cameras	Computers
Japan	960	320
US	320	960

- Japan and US trades to have same number of goods for the products they do not have advantage. . .
  - Japan imports 80 cameras from US
  - US imports 80 computers from Japan

	Cameras	Computers
Japan	880	400
US	400	880

Comparative Advantage

- By specializing and trading, states and individuals can increase overall consumption and efficiency.
- Contradicts the realist view that international affairs are a zero-sum game.
- This also works in the case of 'comparative advantage'

	Cameras	Computers
Japan	9	3
US	4	2

- Production before specialization

	Productivity		Production before specialization	
	Cameras	Computers	Cameras	Computers
Japan	9	3	900	300
US	4	2	400	200

- Specialization (hrs)

	Cameras	Computers
Japan	110	90
US	80	120

- Production after specialization

	Cameras	Computers
Japan	990	270
US	320	240

	Work Productivity per Hours		Before Specialization		Specialization, No Trade		Specialization, with Trade	
Country	Cameras	Computers	Cameras	Computers	Cameras	Computers	Cameras	Computers
Japan	9	3	900	300	990	270	910	300
United States	4	2	400	200	320	240	400	210

- Both benefits . . .

‘What if’s . . .

Change specialization to B), C) Change productivity to 2), 3)

Find win-win situation for 2) and 3)

Productivity

1)	Cameras	Computers
Japan	9	3
US	4	2

2)	Cameras	Computers
Japan	9	3
US	5	2

3)	Cameras	Computers
Japan	6	4
US	4	3

Specialization (hrs)

A)	Cameras	Computers
Japan	110	90
US	80	120

B)	Cameras	Computers
Japan	110	90
US	90	110

C)	Cameras	Computers
Japan	120	80
US	90	110

Implications

- With free trade market will eventually find the equilibrium
- Everyone benefits from unfettered exchanges
  - Open markets and free trade can lift the poor from poverty and expand political liberties
  - Extreme poverty in the world not a result of free trade, but a result of not enough free trade
- Does not guarantee that all will benefit equally

H-O model

- Hecksher-Ohlin (H-O) model
  - Comparative advantage arises from differences in countries' factor endowments.
  - Different factor endowments shape the cost of production (or face different factor prices)
  - Countries have
    - comparative advantage in goods produced using a lot of abundant factor
    - comparative disadvantage in goods produced using a lot of scarce factor

Politics of Trade Cooperation

Why do we still hesitate to trade?

- Are you going to be happy . . .
  - to lose your job because you are in a comparatively disadvantaged sector?
  - to specialize in low-value added goods?
  - when new technology changes the world and you do not have the technology because your country specialised in low-tech industry?

- Trade logic
  - Import – good for welfare of consumers
  - Export – inevitable price to pay for imports
  - Biased towards consumers

- Government
  - Imports – expenditure
  - Exports – income
  - Biased towards producers

- In general, governments strive to . . .
  - Open foreign markets so that their competitive industries can increase their exports
  - Protect less competitive industries from imports

  - Necessitates reciprocal agreements
  - → enforcement problem

Enforcing Agreement

		United States	
		Liberalize	Protect
China	Liberalize	*L,L* *I*	*L,P* *II*
	Protect	*P,L* *IV*	*P,P* *III*

Preference Orders:
China: *P,L* > *L,L* > *P,P* > *L,P*
United States: *L,P* > *L,L* > *P,P* > *P,L*

Figure 3.5 The Prisoners' Dilemma and Trade Liberalization.

- Enforcement problem
  - Governments cannot be certain that other governments will comply with the trade agreements that they conclude.
  - Reluctant to enter into trade agreements
- Prisoners dilemma
  - Both will play protect
- Pareto suboptimal
  - Pareto optimal – no single actor can be made better off without at the same time making another actor worse off.
  - Pareto suboptimal – outcomes in which it is possible for at least one actor to improve its position without any other actor being made worse off
  - PP produces suboptimal outcome for both
- Nash equilibrium
  - An outcome at which neither player has an incentive to change strategies unilaterally
  - China does not want to shift to L, US does not want to shift to L unilaterally

WTO and Trade Cooperation

- Conditions for cooperation
  - Iterated prisoners dilemma – a game played repeatedly by the same governments
  - Must use reciprocity strategy – to allow governments to reward each other for cooperation
  - Governments must care about the payoffs they will receive in the future game.

- WTO provides first two of the three conditions
  - Creating expectations of repeated interaction through regular interaction
  - Provide information that governments need in order to use reciprocity strategies
    - By collecting and disseminating information
    - Improve transparency

- Dispute Settlement Mechanism